Thermodynamic Stability Constraints on Persistent Open Quantum Subsystems: Control Algebra, Physical Actuation and Energy Cost

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What can make the structure of physical control agree with the structure of propagation? Persistence alone does not select a Clifford algebra. A more specific condition does: for a linear Hermitian control family coupled to one centered, independent field input, switching work is independent of the unknown controlled state exactly when the generators square to a scalar quadratic form. Polarization then gives Clifford relations. If the same generators enter a calibrated propagation symbol, the measured work form is a positive pullback of its spatial shell, not the indefinite spacetime form. This conditional connection separates a physical restriction from a choice of notation.

We realize the coupling with a degenerate carrier and a free chiral field. The same interaction determines the control channel, its error and its added stress. An accurate bit flip requires drive occupation of order $\epsilon^{-1/2}$; squeezed inputs nearly attain the bound, while classical inputs need order ϵ^{-1}. A feedback task with scheduled fresh independent drives joins this actuator to calibration records and has an exact stationary loss and a stability proof. A separate sharp energy bound prices translation-only response readout. For inquiry while quantum content is held, a coherent echo and finite Gibbs bank provide explicit protection, reading and renewal operations. Their clocks, drive preparation and spent outputs remain in the resource account. These finite implementations constrain the control–world interface without deriving the supplied field, propagation time or an indefinitely reusable apparatus.

I. INTRODUCTION

Interaction with a large environment drives a quantum system towards decoherence and thermalization [1, 6]. A subsystem that keeps a non-equilibrium steady state does so through exchanges with its surroundings, which may be regulated by feedback [7–10]. The Q-RAIF question is not only what such regulation costs, but why its control structure should bear any relation to the structure in which the subsystem propagates.

Two distinctions are needed. Positivity of a gradient descent metric does not select a Clifford algebra, and no change of coordinates turns a positive work form into a Lorentzian form. Standard dissipative and coherent controls can stabilize states without a Clifford presentation [15–18]. A connection has to come from an actual interaction and from an operational condition on its use.

Here one field coupling supplies both the operation and its energetic response. For a fixed linear family of Hermitian control generators, a centered independent field input gives a work proportional to the expectation of the generator's square. Requiring that charge to be the same for every unknown carrier state is equivalent to scalar-square, hence Clifford, relations (Proposition 6). This is a condition on a specified calibration experiment, not a consequence of persistence or a claim that all useful controls must satisfy it. A one-state average can conceal its failure. When these generators are also the calibrated generators of propagation, Corollary 7 relates the work form to the spatial propagation shell. Evolution time and the physical alignment of the two uses remain inputs.

The finite actuator also makes the cost of maintaining an interface explicit. A degenerate bit coupled to a free chiral field has a flip channel, outgoing stress and switching work determined by the same unitary. Theorem 8 bounds the drive occupation needed for a given error; squeezed inputs nearly attain it, whereas classical drives do not. Theorem 13 composes this imperfect actuator with bath noise and calibration, and Proposition 14 gives one feasibility condition for records and drive energy. Translation-only response readout has its own sharp mean-energy bound, including retained ancillas and joint decoding (Theorem 16).

An interface can be queried while it carries unknown quantum content. A coherent controlled-parity echo protects that content; a mixed port produces a binary record, and a finite Gibbs bank renews the active oscillator while retaining its spent outputs. Collection loss, control and readout have separate charges. The detailed resonant, packet and finite-band collection constructions are grouped in Appendix A: they support the required access rather than define the main control–propagation question.

The outgoing stress is the matter source used in HAFF-D [29]; the feedback task realizes the supplied flip of the Imprint Chain [36]. These are interfaces between stated models. The chiral field, spatial propagation carrier and external clock are not generated by the persistence calculation. Sections II–III state its algebraic obligations; Sections IV–V establish the physical coupling and the conditional bridge; the energy, persistence and inquiry results then describe what that interface can do at finite resources.

II. PERSISTENCE AS A FEEDBACK PROBLEM

Consider $\mathcal{H}_{\text{tot}} = \mathcal{H}_R \otimes \mathcal{H}_E$ with reduced dynamics

$$\dot{\rho}_R = -i[H_{\text{eff}}, \rho_R] + \mathcal{D}[\rho_R]. \tag{1}$$

The controls must be specified in the composed system dynamics, rather than inferred from a formal stationarity condition on a protocol cost.

Standard quantum control writes $H_{\text{ctrl}} = \sum_k u_k(t) G_k$ with generators of a Lie algebra $\mathfrak{g}$, $[G_i, G_j] = i f_{ijk} G_k$ [14]. The Lie algebra fixes the accessible directions. The size of a correction is quantified by a positive cost, norm or information metric supplied separately. This separation is standard and is not a defect of Lie-algebraic control. Section IV shows where such a cost can come from physically: from the stress of the field that implements the control.

III. WHAT THE OPERATOR ALGEBRA MUST SUPPLY

A. Associativity as a boundary of the model

Remark 1 (Associativity as a control-model boundary)**.** *The master-equation model represents sequential controls by linear maps under ordinary composition, which is associative by definition [1]. This is a declared scope condition, not a theorem excluding every non-associative dynamics. Jordan or octonionic models would need additional bracketing, subsystem and composition rules before a control calculation could be transferred [12, 13]. Q-RAIF Paper A [30] uses the same operator-composition boundary for propagation.*

B. A positive metric for Lyapunov descent

For an auxiliary gradient controller on parameters λ consider the descent rule

$$\dot{\lambda} = -\Gamma\, G^{-1} \nabla_\lambda V, \qquad \Gamma > 0. \tag{2}$$

Proposition 2 (Positive descent metric)**.** *The controller contribution to $\dot{V}$ under Eq. (2) is $-\Gamma(\nabla_\lambda V)^T G^{-1} (\nabla_\lambda V)$. It is nonpositive for every gradient exactly when G^{-1}, equivalently the invertible symmetric matrix G, is positive definite. If G has a negative direction, a gradient along it makes the contribution positive.*

Proof. The first statement is the chain rule. A symmetric quadratic form is nonnegative on all vectors precisely when it is positive semidefinite, and an invertible positive semidefinite matrix is positive definite. □

The descent mechanism therefore selects positive definiteness, not Lorentzian signature. Channel types such as unitary, dephasing and dissipative can be labelled by a discrete index, a direct-sum decomposition or supplied weights; their qualitative difference does not need a negative norm.

Parameter descent is not yet the physical state equation. With drift H_0, dissipator $\mathcal{D}$ and controls $H_{\text{ctrl}}(\lambda) = \sum_i u_i(\lambda) G_i$, the closed loop is

$$\dot{\rho} = -i[H_0 + H_{\text{ctrl}}(\lambda), \rho] + \mathcal{D}[\rho], \qquad \dot{\lambda} = -\Gamma G^{-1} \nabla_\lambda V, \tag{3}$$

and for differentiable $V(\rho, \lambda)$

$$\dot{V} = D_\rho V[-i[H_0 + H_{\text{ctrl}}, \rho] + \mathcal{D}[\rho]] - \Gamma(\nabla_\lambda V)^T G^{-1} (\nabla_\lambda V). \tag{4}$$

A stability proof must control the first term as well. Setting a control amplitude equal to a gradient component gives no general sign.

Dissipative and coherent control can stabilize target states with positive costs and Lie or operator-algebraic controls [15–18]. Von Neumann and C^*-algebras are compatible with this model; neither supplies a preferred quadratic form without a dynamics. These examples also show why persistence alone imposes no indefinite Clifford structure.

C. Clifford presentation and a physical criterion

If one chooses a quadratic space (V, q) and requires control generators to satisfy $v^2 = q(v)\mathbf{1}$, the universal property identifies $Cl(V, q)$ as the minimal algebra for that presentation [11]. This is a definitional fact. It does not show that persistence, channel discrimination or Lyapunov descent imposes those relations. Pauli controls of a qubit satisfy $\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbf{1}$, the relations of $Cl(3, 0)$, so the Clifford package is a convenient description of qubit controls. A particular feedback model may favour some representation because of its controls and costs, but that preference has to be shown in its closed-loop dynamics. Proposition 6 below gives a different, explicit criterion: the centered-input switching charge must be independent of the unknown carrier state for every command. That requirement is stronger than positivity or persistence.

a. State stability. Equation (4) is the continuous-time obligation: a parameter descent controls only its own contribution, while drift, dissipator and controlled state evolution remain. For full-rank states and a fixed full-rank target, $\dot{D}(\rho\|\rho_*) = \mathrm{tr}[\dot{\rho}(\log\rho - \log\rho_*)]$. Theorem 13 establishes stability instead for a discrete composed channel with a physically realized, imperfect correction.

IV. A CONTROL FLIP REALIZED BY A FIELD COUPLING

A. The coupling and its channel

The field-coupling and drive bounds in this and the next section use $\hbar = 1$: $\bar{\omega}N$ and $\bar{\omega}\lambda^2$ are energies in these units. In physical units they are $\hbar\bar{\omega}N$ and $\hbar\bar{\omega}\lambda^2$; the field Hamiltonian and stress carry the same overall $\hbar$. Quadratures, occupation and pulse areas are dimensionless. The later delay, entropy and retained-oscillator accounts display $\hbar$ explicitly.

The controlled system is a two-level system with zero free Hamiltonian, Pauli operators X and Z. A bosonic mode with $[b, b^\dagger] = 1$ and quadratures $Q = b + b^\dagger$, $P = i(b^\dagger - b)$, $[Q, P] = 2i$, interacts through

$$U = \exp[-i\lambda XQ], \qquad \lambda > 0. \tag{5}$$

In the eigenbasis $X|x\rangle = x|x\rangle$, $U = \sum_x |x\rangle\langle x| \otimes e^{-i\lambda xQ}$, and $e^{-i\lambda xQ}$ is the displacement $D(-i\lambda x)$. Write $\chi_\sigma(k) = \mathrm{tr}(\sigma e^{-ikQ})$ for the characteristic function of Q in a mode state σ.

Proposition 3 (Channel of the coupling)**.** *Let the mode start in a state σ independent of the bit. The bit channel multiplies the matrix element $\rho_{xx'}$ in the X basis by $\chi_\sigma(\lambda(x - x'))$. For bit states diagonal in Z, its output Z statistics agree with a flip having failure probability*

$$\epsilon = \tfrac{1}{2}\big(1 + \mathrm{Re}\,\chi_\sigma(2\lambda)\big). \tag{6}$$

For a coherent input $|\alpha\rangle$, $\alpha > 0$, the factor is

$$e^{-\lambda^2(1-xx')-2i\alpha\lambda(x-x')}. \tag{7}$$

If $2\lambda\alpha = \pi/2$ and $N = \alpha^2$, the channel on every bit state is

$$\mathcal{F}_N(\rho) = s_N X\rho X + (1 - s_N)\rho, \qquad s_N = \frac{1 + e^{-2\lambda^2}}{2}, \qquad \lambda^2 = \frac{\pi^2}{16N}, \tag{8}$$

so that $\epsilon_N = 1 - s_N = \big(1 - e^{-\pi^2/(8N)}\big)/2$.

Proof. Tracing the mode gives the factor $\mathrm{tr}(\sigma\, e^{i\lambda x'Q}e^{-i\lambda xQ}) = \chi_\sigma(\lambda(x - x'))$. A Z-diagonal state $(I + zZ)/2$ has $\rho_{+-} = z/2$ in the X basis, so the output mean of Z is $z\,\mathrm{Re}\,\chi_\sigma(2\lambda)$, which is $(1-2s)z$ for a flip with success probability s. For the coherent input, $D(\beta)D(\alpha) = e^{(\beta\alpha^* - \beta^*\alpha)/2}D(\alpha + \beta)$ gives $e^{-i\lambda xQ}|\alpha\rangle = e^{-i\alpha\lambda x}|\alpha - i\lambda x\rangle$. The overlap is the coherent-state inner product $\langle\alpha - i\lambda x'|\alpha - i\lambda x\rangle = e^{-\lambda^2(1-xx')-i\alpha\lambda(x-x')}$ times the Weyl phase $e^{-i\alpha\lambda(x-x')}$ from composing the displacements; this is Eq. (7). At $2\lambda\alpha = \pi/2$ the off-diagonal factor is $-e^{-2\lambda^2}$, and $1 - 2s_N = -e^{-2\lambda^2}$ yields Eq. (8) on all inputs. □

The channel $\mathcal{F}_N$ differs from the ideal Ad_X by $\epsilon_N(\mathrm{id} - \mathrm{Ad}_X)$, so its half diamond-norm distance from the ideal flip is ϵ_N, and its average gate infidelity is $2\epsilon_N/3$. These are different error conventions. The coherent family above needs infinite occupation for an exact flip; Section VI shows that this holds for every input state.

B. A local switch on a free field

Equation (5) becomes a local operation once the mode is defined by a switch. Take a right-moving free massless scalar in $1+1$ dimensions with null coordinate u,

$$\phi(u) = \int_0^\infty \frac{d\omega}{\sqrt{4\pi\omega}} \big(a_\omega e^{-i\omega u} + a_\omega^\dagger e^{i\omega u}\big), \qquad [a_\omega, a_{\omega'}^\dagger] = \delta(\omega - \omega'), \tag{9}$$

current $j = \partial_u \phi$ and stress $T_{uu} =: j^2 :$. The current obeys $[j(u), j(v)] = \frac{i}{2}\delta'(u-v)$. For a real smooth compactly supported switch g define

$$Q_g = \int g(u)\, j(u)\, du = \lambda(b + b^\dagger), \qquad \lambda^2 = \int_0^\infty \frac{\omega}{4\pi} |\hat{g}(\omega)|^2 d\omega, \qquad \hat{g}(\omega) = \int g(u) e^{-i\omega u} du, \tag{10}$$

where $b = \int_0^\infty f^*(\omega) a_\omega d\omega$ with $f(\omega) = i\sqrt{\omega/(4\pi)}\, \hat{g}^*(\omega)/\lambda$ is a normalized mode. Its mean frequency is $\bar{\omega} = \int \omega |f|^2 d\omega$, and the current contains it as $j(u) = \psi(u) b + \psi^*(u) b^\dagger + j_\perp(u)$ with $\psi(u) = -i \int_0^\infty \sqrt{\omega/(4\pi)}\, f(\omega) e^{-i\omega u} d\omega$.

A probe at fixed position with interaction Hamiltonian $g(t) X j(t)$ evolves by $U = e^{i\vartheta} \exp(-iXQ_g)$ for a real phase ϑ. The reason is that $[g(t)Xj(t), g(t')Xj(t')]$ is a multiple of the identity, because $X^2 = 1$ and the current commutator is a c-number, so the Magnus series stops at second order and the second-order term is a phase. This uses three conditions: a single probe, a fixed coupling operator X, and a probe free Hamiltonian that is zero or commutes with X. Nonzero field frequencies are harmless. A probe Hamiltonian that does not commute with X makes $X(t)$ time dependent, and several probes keep relative phases; neither case is covered. A point-like limit of the switch needs regularization, so we use smooth profiles. This exact Weyl evolution of a degenerate detector is a known method [21, 22]; what we add is its use as a priced feedback actuator.

C. Stress, work and a derived quadratic cost

Proposition 4 (Stress of one encounter)**.** *For the unitary $U = \exp(-iXQ_g)$,*

$$U^\dagger j(u) U = j(u) + \tfrac{1}{2} X g'(u), \qquad U^\dagger T_{uu}(u) U = T_{uu}(u) + g'(u)\, Xj(u) + \tfrac{1}{4} g'(u)^2. \tag{11}$$

The expected added stress is $g'(u)\langle Xj(u)\rangle + g'(u)^2/4$. The cross term vanishes if the bit and the fresh field input are initially independent and $\mathrm{tr}(\rho X) = 0$, or if their joint state has no coherence in the bit's Z basis. A Z-diagonal bit marginal alone is not sufficient.

Proof. $[Q_g, j(u)] = \int g(v)[j(v), j(u)] dv = -\frac{i}{2} g'(u)$ is a c-number, so $e^{iXQ_g} j(u) e^{-iXQ_g} = j(u) + \frac{1}{2} X g'(u)$ exactly. Squaring, using $X^2 = 1$, and noting that a c-number shift leaves normal ordering unchanged gives the stress identity. Under independence $\langle Xj\rangle = \mathrm{tr}(\rho X)\langle j\rangle = 0$, even when the input beam has $\langle j\rangle \neq 0$. If the joint state is block diagonal in the bit's Z basis, $X \otimes A$ has zero expectation for every field operator A. For the last claim, $(|{\uparrow}\rangle|0\rangle + |{\downarrow}\rangle|1_b\rangle)/\sqrt{2}$ has bit marginal $I/2$ but $\langle Xb\rangle = 1/2$, hence $\langle Xj(u)\rangle = \mathrm{Re}\,\psi(u) \neq 0$; with a relative phase i the integrated cross term equals $-\lambda\bar{\omega}$ and changes the work below. □

Under the stated joint condition the added stress is

$$F_g(u) = \tfrac{1}{4} g'(u)^2 = 4\lambda^2 [\mathrm{Im}\,\psi(u)]^2, \tag{12}$$

the second form following from $2\lambda\,\mathrm{Im}\,\psi = g'/2$. If the interaction is off before and after the encounter and the bit Hamiltonian is zero, the energy delivered by the external switching agent equals the change of field energy,

$$W_{\mathrm{switch}} = \frac{1}{4} \int g'(u)^2 du = \frac{1}{4\pi} \int_0^\infty \omega^2 |\hat{g}(\omega)|^2 d\omega = \lambda^2 \bar{\omega}, \tag{13}$$

by Parseval's identity and $|f|^2 = \omega |\hat{g}|^2/(4\pi\lambda^2)$. This holds for every input state of the field that satisfies the joint condition.

Proposition 5 (Derived quadratic cost)**.** *For $g = \sum_i a_i g_i$ with real smooth compactly supported g_i, $W_{\mathrm{switch}} = \sum_{ij} a_i K_{ij} a_j$ with $K_{ij} = \frac{1}{4}\int g_i' g_j' du$. If the g_i are linearly independent, K is positive definite.*

Proof. $a^T K a = \frac{1}{4}\int (\sum_i a_i g_i')^2 du$ vanishes only if $\sum_i a_i g_i$ is constant; compact support makes the constant zero, and linear independence gives $a = 0$. □

D. Coherent drive

Let the mode b carry a coherent state $|\alpha\rangle$, $\alpha = \sqrt{N}$, with all orthogonal modes in vacuum, independent of a Z-diagonal bit. The output field marginal is then

$$\rho_F^{\text{out}} = \tfrac{1}{2}|\alpha - i\lambda\rangle\langle\alpha - i\lambda| + \tfrac{1}{2}|\alpha + i\lambda\rangle\langle\alpha + i\lambda|, \tag{14}$$

independently of the bit's bias, although bit and field are correlated afterwards. The incoming field energy is $E_{\text{in}} = N\bar{\omega}$ and the outgoing energy is $(N+\lambda^2)\bar{\omega}$. A wave packet is not an eigenmode of the field Hamiltonian, so these are mean values; the field Hamiltonian is not $\bar{\omega}b^\dagger b$. The incoming stress is $4N[\operatorname{Re}\psi(u)]^2$ and the encounter adds F_g of Eq. (12). The gravitational use of these profiles is developed in Ref. [29].

V. CONTROL STRUCTURE AND PROPAGATION

A. State-independent charging selects scalar squares

The fixed Pauli coupling can be generalized without changing the field. Let $A(a) = \sum_i a_i A_i$ be a real linear family of dimensionless Hermitian operators on a finite carrier. During one encounter a is fixed, and the carrier free Hamiltonian vanishes or commutes with $A(a)$. The switch g is fixed, nonzero, smooth and compactly supported. The input field is independent of the carrier and centered, $\langle j(u)\rangle = 0$, with finite energy; the vacuum is one example. Put $\kappa_g = \frac{1}{4}\int g'(u)^2 du > 0$, in the $\hbar = 1$ convention of Section IV.

Proposition 6 (Content-independent switching work)**.** *For the interaction $g(t)A(a)j(t)$ the field shift and switching work are*

$$U^\dagger j(u) U = j(u) + \tfrac{1}{2}A(a)g'(u), \tag{15}$$

$$W_\rho(a) = \kappa_g \operatorname{tr}\big[\rho A(a)^2\big]. \tag{16}$$

The following conditions are equivalent:

1. *$W_\rho(a)$ is independent of the carrier density matrix ρ for every command a;*
2. *$A(a)^2 = Q_{\text{ctrl}}(a)I$ for every a;*
3. *for a real symmetric matrix Q_{ctrl}, $\{A_i, A_j\} = 2(Q_{\text{ctrl}})_{ij}I$.*

The form is positive semidefinite, and positive definite if $a \mapsto A(a)$ is injective. In that case the generators give a representation of its Euclidean Clifford algebra.

Proof. The field commutator is a c-number. The second Magnus term is $i\vartheta A(a)^2$ and all higher terms vanish, since $A(a)^2$ commutes with $A(a)$ and with the field. Thus $U = e^{i\vartheta A(a)^2}e^{-iA(a)Q_g}$. The first factor need not be a global phase, but it commutes with field observables. The Weyl commutator calculation of Proposition 4 gives the stated shift and

$$U^\dagger T_{uu} U - T_{uu} = g'(u)A(a)j(u) + \tfrac{1}{4}g'(u)^2 A(a)^2.$$

Independence and centering remove the cross term for every ρ. The carrier free energy is unchanged and the interaction is off at the endpoints, so integration gives Eq. (16). A Hermitian operator has the same expectation in every density matrix exactly when it is a scalar multiple of I. This proves the first equivalence. Expanding $A(a)^2$ and comparing polynomial coefficients gives the second. Finally $Q_{\text{ctrl}}(a)I = A(a)^2 \geq 0$, and equality implies $A(a) = 0$. □

The all-state condition matters. For $A(a) = a_1 Z + a_2 I$ a maximally mixed carrier has work $\kappa_g(a_1^2 + a_2^2)$, although $A(a)^2 = (a_1^2 + a_2^2)I + 2a_1 a_2 Z$ is not scalar. The two Z eigenstates instead cost $\kappa_g(a_1 \pm a_2)^2$. Thus a positive average charge does not impose Clifford relations. Correlated inputs or a noncentered field introduce the cross term in the proof and need their own account. In particular this centered-input calibration is not a claim of state-independent work for every coherent drive used to rotate the carrier; nor does it guarantee a useful unitary gate.

B. A calibrated bridge to the spatial shell

Write a propagation symbol on the same internal carrier as $J(p) = \sum_i p_i \alpha_i$, with linearly independent Hermitian α_i and $J(p)^2 = Q_{\mathrm{prop}}(p) I$. The scalar-square argument makes Q_{prop} positive definite. Here J has energy units and the spatial carrier is supplied. Let C be a calibrated linear map from commands to momenta and $E_* > 0$ a fixed energy scale. Coupling the actual control through $A(a) = J(Ca)/E_*$ is a physical alignment condition, not a relabelling of two unrelated quadratic forms.

Corollary 7 (Aligned work and propagation)**.** *Under the encounter assumptions of Proposition 6 and this alignment,*

$$W(a) = a^T K_{\mathrm{op}} a, \qquad K_{\mathrm{op}} = \frac{\kappa_g}{E_*^2}\, C^T Q_{\mathrm{prop}} C. \tag{17}$$

Conversely, if C is invertible and this aligned family satisfies the all-state charge condition, it imposes a scalar quadratic shell on J. If in addition a supplied mass generator satisfies $\beta^2 = I$ and $\{\beta, \alpha_i\} = 0$, then

$$[\,\omega\beta - \beta J(p)\,]^2 = [\,\omega^2 - Q_{\mathrm{prop}}(p)\,] I. \tag{18}$$

Proof. Substitution in Eq. (16) proves the work identity. An invertible C ranges over all momenta, so the converse is the scalar-square criterion. For the last identity, $\beta J = -J\beta$ cancels the cross terms and $(\beta J)^2 = -J^2$. □

In Eq. (18), ω denotes energy conjugate to the independently supplied time, with $\hbar = 1$. The relative sign comes from propagation and the mass anticommutator, not from turning a positive work metric into an indefinite one. This is the closed-carrier symbol map of Q-RAIF C [31] with an explicit calibrated cost attached to it.

A nonempty case uses $\alpha_1 = X$, $\alpha_2 = Z$ and $\beta = Y$. In units $E_* = 1$, $Q_{\mathrm{prop}}(p) = p_1^2 + p_2^2$. For $C = \mathrm{diag}(1, 2)$, $K_{\mathrm{op}}/\kappa_g = \mathrm{diag}(1, 4)$ while the propagation shell is unchanged. The unknown calibration cannot be inferred from positivity of the cost. During the encounter the carrier must still obey the commuting-free-Hamiltonian condition. A general massive propagation Hamiltonian does not commute with $A(a)$; pausing or compensating it requires physical control and a separate cost. The example therefore supplies two compatible, calibrated uses of the carrier, not an autonomous spacetime-generating device.

C. Work, descent and spacetime

The descent metric G in Proposition 2 prices changes of controller parameters. Proposition 5 derives a work form K on switch profiles; Eq. (17) instead prices operator amplitudes for one fixed switch. These are different ways to vary a coupling. If controller parameters are the amplitudes being priced and the corresponding work form is positive definite, choosing it as their descent metric makes $-K^{-1}\nabla V$ the steepest direction at fixed quadratic charge, by Cauchy–Schwarz in the K inner product. That is a design choice; the feedback of Section VII has a stability proof without any gradient metric.

The new restriction is not equality of all three forms. It is the testable relation (17), under known coupling, calibration and input conditions. Its failure rejects that aligned model. Without those conditions, persistence supplies no route from a positive cost to a spatial shell or to Lorentzian signature.

VI. THE ENERGY COST OF AN ACCURATE FLIP

How much drive is needed for a flip of given accuracy? The coherent family gives $N = \pi^2/(16\ell)$ with $\ell = -\frac{1}{2}\log(1 - 2\epsilon)$, about $\pi^2/(16\epsilon)$. The following theorem bounds every input state.

Theorem 8 (Energy–accuracy bound)**.** *Let σ be any state of the mode b with finite mean occupation $N = \mathrm{tr}(\sigma b^\dagger b)$, let $\lambda > 0$, and let the coupling* (5) *flip Z-diagonal bit states with failure probability $\epsilon \in (0, \frac{1}{2})$ as in Eq.* (6)*. With $\theta_\epsilon = \arccos(2\epsilon - 1) \in (\pi/2, \pi)$,*

$$N \geq \frac{\theta_\epsilon(1 - 2\epsilon)}{4\sqrt{\epsilon(1 - \epsilon)}} - \frac{1}{2}, \tag{19}$$

$$N + \lambda^2 \geq \frac{\theta_\epsilon}{4\sqrt{\epsilon(1 - \epsilon)}} - \frac{1}{2}. \tag{20}$$

Both right-hand sides equal $\pi/(4\sqrt{\epsilon}) - 1 + O(\sqrt{\epsilon})$. The second inequality is a bound on an occupation-plus-activation budget. It equals the outgoing single-mode energy in units of $\bar{\omega}$ only with the independent zero-cross-term field inputs of Section IV C; correlations can change the actual switching work. An exact flip, $\epsilon = 0$, is impossible at finite occupation.

Proof. Write $C = \cos(2\lambda Q)$ and $U_2 = e^{-2i\lambda Q}$, so that $\langle C\rangle = \operatorname{Re}\chi_\sigma(2\lambda) = 2\epsilon - 1$ and $b^\dagger b = (Q^2 + P^2 - 2)/4$. Finite N makes both second moments finite.

Position quadrature. The function $G(c) = \arccos^2 c$ is convex on $[-1, 1]$: with $c = \cos y$, $y \in (0, \pi)$, one finds $G''(c) = 2(1 - y\cot y)/\sin^2 y \geq 0$, since $y\cot y \leq 1$. For every real q, $(2\lambda q)^2 \geq G(\cos 2\lambda q)$, with equality when $|2\lambda q| \leq \pi$ and strict inequality otherwise because $G \leq \pi^2$. Jensen's inequality for the spectral distribution of Q gives

$$\langle Q^2\rangle \geq \frac{G(\langle C\rangle)}{4\lambda^2} = \frac{\theta_\epsilon^2}{4\lambda^2}. \tag{21}$$

Momentum quadrature. Since $U_2^\dagger P U_2 = P - 4\lambda$, we have $[U_2, P] = 4\lambda U_2$ on the domain of P. Writing $\delta U_2 = U_2 - \langle U_2\rangle$, $\delta P = P - \langle P\rangle$ and applying Cauchy–Schwarz to the Hilbert–Schmidt products $\operatorname{tr}(\sigma\,\delta U_2\,\delta P)$ and $\operatorname{tr}(\sigma\,\delta P\,\delta U_2)$ gives $|\langle [U_2, P]\rangle| \leq 2\,\Delta U_2\,\Delta P$ with $(\Delta U_2)^2 = 1 - |\langle U_2\rangle|^2$. Hence $\Delta P \geq 2\lambda|\chi|/\sqrt{1 - |\chi|^2}$ with $\chi = \chi_\sigma(2\lambda)$. Because $|\chi| \geq |\operatorname{Re}\chi| = 1 - 2\epsilon$ and $x/\sqrt{1 - x^2}$ increases,

$$\langle P^2\rangle \geq (\Delta P)^2 \geq \frac{\lambda^2(1 - 2\epsilon)^2}{\epsilon(1 - \epsilon)}. \tag{22}$$

In particular $|\chi| < 1$ at finite N, which excludes $\epsilon = 0$.

Combination. Adding the two bounds,

$$N \geq \frac{1}{4}\left[\frac{\theta_\epsilon^2}{4\lambda^2} + \frac{\lambda^2(1 - 2\epsilon)^2}{\epsilon(1 - \epsilon)}\right] - \frac{1}{2}, \tag{23}$$

and minimizing over λ by the arithmetic–geometric mean inequality gives Eq. (19). Adding λ^2 changes the coefficient of λ^2 to $\frac{1}{4}[(1 - 2\epsilon)^2/(\epsilon(1 - \epsilon)) + 4] = 1/[4\epsilon(1 - \epsilon)]$, and the same step gives Eq. (20). The expansion uses $\theta_\epsilon = \pi - 2\sqrt{\epsilon} + O(\epsilon^{3/2})$. □

The bound is nearly attained by Gaussian drives.

Proposition 9 (Coherent and Gaussian families)**.** *(a) The coherent family with $2\lambda\alpha = \pi/2$ needs $N = \pi^2/(16\ell)$, $\ell = -\frac{1}{2}\log(1 - 2\epsilon)$. (b) For a displaced pure Gaussian state whose Q has mean 2α and variance $V > 0$, $\chi(2\lambda) = e^{-4i\lambda\alpha}e^{-2\lambda^2 V}$ and $N = \alpha^2 + (V + V^{-1} - 2)/4$. At $2\lambda\alpha = \pi/2$ and failure ϵ,*

$$\min N = \frac{\sqrt{1 + \pi^2/(4\ell)} - 1}{2}, \qquad \min(N + \lambda^2) = \frac{1}{2}\sqrt{\Big(1 + \frac{\pi^2}{4\ell}\Big)(1 + 4\ell)} - \frac{1}{2}, \tag{24}$$

attained at $V = (1 + \pi^2/(4\ell))^{-1/2}$ and $V = [(1 + 4\ell)/(1 + \pi^2/(4\ell))]^{1/2}$ respectively. Both equal $\pi/(4\sqrt{\epsilon}) - \frac{1}{2} + O(\sqrt{\epsilon})$, so their gap to Theorem 8 tends to half a quantum.

Proof. The Gaussian characteristic function gives χ. The failure condition reads $\lambda^2 V = \ell$, so $\alpha^2 = \pi^2/(16\lambda^2) = aV$ with $a = \pi^2/(16\ell)$ and $N = (1 + 4a)V/4 + 1/(4V) - \frac{1}{2}$, minimized at $V = (1 + 4a)^{-1/2}$. Adding $\lambda^2 = \ell/V$ replaces $1/(4V)$ by $(1 + 4\ell)/(4V)$. □

Squeezing is what makes the better scaling possible.

Proposition 10 (Classical drives)**.** *If $\sigma = \int P(\beta)|\beta\rangle\langle\beta|\,d^2\beta$ with a probability density $P \geq 0$, then a flip with failure ϵ needs*

$$N \geq \frac{(\pi - \arccos(1 - 2\epsilon))^2}{16\ell} = \frac{\pi^2}{16\epsilon}\big(1 - 4\sqrt{\epsilon}/\pi + O(\epsilon)\big). \tag{25}$$

Proof. Each coherent component has $\chi_\beta(2\lambda) = e^{-2\lambda^2}e^{-4i\lambda\operatorname{Re}\beta}$, so $\int P\cos(4\lambda\operatorname{Re}\beta) = -(1 - 2\epsilon)e^{2\lambda^2}$. This must be at least -1, hence $\lambda^2 \leq \ell$. The convexity argument of Theorem 8 applied to $y = 4\lambda\operatorname{Re}\beta$ gives $16\lambda^2\int P(\operatorname{Re}\beta)^2 \geq (\pi - \arccos[(1 - 2\epsilon)e^{2\lambda^2}])^2 \geq (\pi - \arccos(1 - 2\epsilon))^2$. Finally $N = \int P|\beta|^2 \geq \int P(\operatorname{Re}\beta)^2$ and $\lambda^2 \leq \ell$. □

ϵ	Classical bound (25)	Coherent, $2\lambda\alpha = \pi/2$	Bound (19)	Grid variational	Gaussian half-turn (24)
0.1	3.496	5.529	1.165	1.847	1.904
0.05	8.589	11.709	2.278	2.921	2.958
0.01	53.53	61.07	6.742	7.314	7.330
0.001	591.7	616.2	23.80	24.32	24.33

TABLE I. Mean drive occupation needed for a flip with failure probability ϵ. Including one activation's switching work at $\epsilon = 0.01$: bound (20) 6.890, grid variational minimum 7.470, Gaussian half-turn 7.487, coherent 61.08. The grid minima change by less than 2×10^{-3} when the spacing is halved; this is not a certified global error bound.

At fixed λ, a constrained minimum can be certified by a supporting ground-state problem. If ψ_μ is the ground state of $b^\dagger b + \mu \cos(2\lambda Q)$ and $\langle C \rangle_{\psi_\mu} = 2\epsilon - 1$, then every σ with the same $\langle C \rangle$ satisfies $\mathrm{tr}(\sigma b^\dagger b) \geq \langle b^\dagger b \rangle_{\psi_\mu}$, because $\mathrm{tr}\, \sigma(b^\dagger b + \mu C)$ is at least the ground energy. Table I lists numerical variational minima on a position grid, further minimized over λ, alongside the analytic bounds and the explicitly attainable Gaussian half-turn family. Halving the grid checks resolution stability, not the continuous global minimum. The computed states are close to displaced squeezed states, while the coherent half-turn family uses substantially more occupation.

Remark 11 (Scope of the occupation)**.** *(i) For a multimode field state, Cauchy–Schwarz applied to $b = \int f^* a_\omega d\omega$ gives $\langle b^\dagger b \rangle \leq \omega_h^{-1} \langle H_F \rangle$ with $\omega_h^{-1} = \int |f|^2 \omega^{-1} d\omega$, so the field energy is at least ω_h times the bound* (19)*; under the zero-cross-term condition of Section IV C, $\omega_h \leq \bar{\omega}$ then makes field energy plus switching work at least ω_h times the bound* (20)*. (ii) The occupation is the drive level present at the encounter, not energy absorbed by the bit. The bit is degenerate; the beam leaves with energy $(N + \lambda^2)\bar{\omega}$ in the coherent case. Whether drive energy is lost depends on whether the outgoing beam is recovered. (iii) Squeezed inputs can have locally negative normal-ordered stress, and their preparation and phase reference are additional resources.*

Recovering the beam is not free of consequences. Two bits placed one after the other on the same chiral packet, with the same switch profile in u, experience $U = \exp[-i\lambda(X_1 + X_2)Q]$.

Proposition 12 (Shared drive correlates errors)**.** *For a shared drive each bit's marginal channel is the single-encounter channel. For two bits starting at $Z = +1$, $\langle Z_1 Z_2 \rangle = \frac{1}{2}[1 + \mathrm{Re}\, \chi(4\lambda)]$. For the coherent family of Proposition 3 this equals $(1 + e^{-8\lambda^2})/2$, which exceeds the product $\langle Z_1 \rangle \langle Z_2 \rangle = e^{-4\lambda^2}$ obtained with independent drives.*

Proof. In the X basis the joint element with indices $(x_1 x_2), (x_1' x_2')$ is multiplied by $\chi(\lambda(x_1 + x_2 - x_1' - x_2'))$; tracing one bit recovers Proposition 3. $Z_1 Z_2$ connects (x_1, x_2) with $(-x_1, -x_2)$, which gives arguments $\pm 4\lambda$ and 0. For the coherent family $\chi(4\lambda) = e^{-8\lambda^2} e^{-2\pi i}$, and $(1 + y^2)/2 > y$ for $0 < y < 1$. □

At $N = 16$ the single failure probability is 0.0371, and the probability that both flips fail is 0.00393, against 0.00138 for independent drives. Drive energy can be shared only at the price of correlated failures. The persistence analysis below uses a fresh drive at each opportunity.

VII. PERSISTENCE WITH A FINITE-ENERGY ACTUATOR

A. The task

We use the feedback task of the Imprint Chain [36] with the corrective flip replaced by the coupling of Section IV. The persistent subsystem is a bit restricted to Z-diagonal states, with value $W = \pm 1$ and target $W = +1$. A bath step flips W with probability $d \in (0, \frac{1}{2})$; put $c = 1 - 2d$. A reading prepares a fresh probe, copies W by a CNOT, rotates the probe by U_a with $a \in \{0, 1\}$, and measures it with effects $E_{y|s} = \frac{1}{2}(I + y\nu\sigma_s)$, $\sigma_0 = Z$, $\sigma_1 = X$, where the instrument label S is hidden and fixed. Then

$$\Pr(Y = y \mid W = w, S = s, a) = \begin{cases} (1 + yw\nu)/2, & a = s, \\ 1/2, & a \neq s. \end{cases} \tag{26}$$

A calibration stage selects a; its mismatch probability $q = \Pr(a \neq S)$ is fixed along a trajectory. In each task step the system reads Y, accepts feedback with an independent probability $h \in (0, 1]$, switches on the coupling if accepted and $Y = -1$, and then the bath acts. The loss is the occupation probability $L_t = \Pr(W_t = -1)$.

Two declarations fix the resource accounting. Each opportunity has its own drive, initially independent of the bit and of earlier records, and the outgoing field never returns to the controller's reading access. The drive is scheduled: the same incoming beam arrives at every opportunity and the command only decides whether g is switched on. On-demand emission of a whole packet would need a router and a timing scheme that we do not model.

B. Closed-loop stability

Theorem 13 (Persistence with a finite-energy actuator)**.** *Let each coupling act on the bit as the flip channel $sX\rho X + (1-s)\rho$, which is the case exactly when $\chi_\sigma(2\lambda)$ is real, for instance for the coherent and Gaussian families at $2\lambda\alpha = \pi/2$. Put $\Lambda = c(1-hs)$ and $B = chs\nu$, and let the task start from W uniform.*

(a) Conditional on $a = S$, the bit evolves by a fixed channel $\mathcal{T}$ on Z-diagonal states, $m \mapsto \Lambda m + B$ for the mean m of W. Conditional on $a \neq S$, $m \mapsto \Lambda m$.

(b) $\mathcal{T}$ has the unique fixed point $\rho_* = (I + m_* Z)/2$, $m_* = B/(1-\Lambda)$. The trace distance to ρ_* contracts by exactly Λ per step, and $D(\rho_t\|\rho_*)$ is nonincreasing and tends to zero.

(c) The stationary loss is

$$L_\infty(q) = \frac{1}{2} - \frac{(1-q)B}{2(1-\Lambda)}. \tag{27}$$

(d) The stationary rate of actual couplings is

$$p(q) = \frac{h}{2}\left[1 - \frac{(1-q)\nu B}{1-\Lambda}\right] = h\nu L_\infty(q) + \frac{h(1-\nu)}{2}, \tag{28}$$

of which a fraction s succeed. Over a window of T steps the expected number of couplings is $\frac{h}{2}\big[T - (1-q)\nu\frac{B}{1-\Lambda}\big(T - \frac{1-\Lambda^T}{1-\Lambda}\big)\big]$.

(e) For two calibrations with $q' < q$, $L_\infty(q) - L_\infty(q') = (q-q')B/[2(1-\Lambda)]$ and $p(q) - p(q') = h\nu[L_\infty(q) - L_\infty(q')]$.

Proof. With $a = S$ a reading has $\mathbb{E}[WY] = \nu$, and the coupling is attempted with probability $h(1-Y)/2$ and flips with probability s. Thus the mean after feedback is $m - 2s\,\mathbb{E}[Wh(1-Y)/2] = (1-hs)m + hs\nu$, and the bath multiplies it by c. With $a \neq S$, Y is independent of W and the mean after feedback is $(1-hs)m$. Fresh probes and fresh drives make each step the same channel on the bit marginal. Trace distance between Z-diagonal states is $|m-m'|/2$, which the affine map multiplies by $\Lambda < 1$. Data processing gives $D(\mathcal{T}\rho\|\rho_*) \leq D(\rho\|\rho_*)$, and $m_t \to m_*$ gives convergence. Starting from $m_0 = 0$, the branch means are $m_*(1-\Lambda^t)$ and 0, weighted by $1-q$ and q; this gives (c). The attempt probability is $h(1-\nu m)/2$ for $a = S$ and $h/2$ otherwise, which gives (d) after averaging and using $(1-q)m_* = 1 - 2L_\infty$. Part (e) follows from (c) and (d). □

Gate failure enters the recursion through hs, while the attempt rate keeps h: a failed gate uses the whole encounter. Stability here does not rely on a gradient rule or a metric G. It follows from the contraction of the composed channel, and the relative entropy to the stationary state is a Lyapunov function of the full closed loop, which is what Section III C 0 a required of a proof. Whether persistence is useful is a separate matter: the stationary polarization is positive only on the calibrated branch and grows with s.

Table II uses $\nu = 0.8$, $c = 0.9$, $h = 0.4$ and coherent drives. The calibrated column uses 32 passive two-read blocks, $q_{32} = 0.0404888$ (Section VII C). With an exact flip the calibrated loss would be 0.199631 [36].

N	ϵ_N	Loss, $q = \frac{1}{2}$	Loss, q_{32}	Rate p, $q = \frac{1}{2}$	Rate p, q_{32}	Added energy per opportunity
1	0.354394	0.360165	0.231654	0.155253	0.114129	0.070401
4	0.132699	0.348517	0.209300	0.151525	0.106976	0.016497
16	0.037104	0.344779	0.202127	0.150329	0.104680	0.004036
64	0.009546	0.343806	0.200259	0.150018	0.104083	0.001003

TABLE II. Stationary loss and coupling rate with a coherent drive of occupation N. The last column is $p\lambda^2$ at q_{32} in units of $\bar{\omega}$; the incoming energy $N\bar{\omega}$ is paid at every opportunity in addition. Reading, probe and calibration costs are not included.

C. Calibration with the same imperfect gate

The Imprint Chain separates calibration blocks by an exact randomization $\mathcal{R}(\rho) = \frac{1}{2}(\rho + X\rho X)$. With an imperfect gate, triggering it with probability $\frac{1}{2}$ leaves a fraction $1 - s$ of the bit's mean. Triggering with probability $1/(2s)$, which is at most one because $s \geq \frac{1}{2}$, and discarding the coin gives

$$\Big(1 - \frac{1}{2s}\Big)\rho + \frac{1}{2s}\big(sX\rho X + (1-s)\rho\big) = \tfrac{1}{2}(\rho + X\rho X) \tag{29}$$

exactly. This presumes that s is known. If the trigger probability is rounded to k binary digits, each randomization has flip probability within $s2^{-k-1}$ of $\frac{1}{2}$. Since $\mathrm{id} - \mathrm{Ad}_X$ has diamond norm 2, the record distributions of a protocol with $n+1$ randomizations differ from the exact ones by at most $(n+1)s2^{-k-1}$ in total variation, or 1.52×10^{-5} for $N = 16$, $n = 32$, $k = 20$.

Passive two-read blocks randomize, read, let the bath act and read again, without a feedback flip. With the compensated randomization they keep the Imprint Chain likelihood: the product of the two readings is positive with probability $p_{\mathrm{cal}} = (1 + c\nu^2)/2$ if $a = S$ and $\frac{1}{2}$ otherwise. After n blocks the optimal mismatch probability is

$$q_n = \frac{1}{2}\sum_{k=0}^{n}\binom{n}{k}\min\big\{p_{\mathrm{cal}}^k(1-p_{\mathrm{cal}})^{n-k}, 2^{-n}\big\}. \tag{30}$$

Active blocks, with a feedback flip after the first reading, change once the flip is imperfect. For $a = S$,

$$\Pr(y_1, y_2) = \tfrac{1}{4}\big[1 + c\nu^2 y_2\big(s + (1-s)y_1\big)\big], \tag{31}$$

and the pair is uniform for $a \neq S$. This law is obtained from the passive pair by an S-independent post-processing that keeps the first reading and relabels with probability s, so gate error creates no calibration information about the sensor. The success probability s itself is readable from the closed loop: in the persistence task the lag-one correlation of readings on gated steps differs from that on ungated steps by a term proportional to s, so the agent's own records determine s, the bath contraction and the squared gain without references, and never the sign [33]. Calibration with n blocks uses $2n$ readings and $n+1$ randomizations with an expected $(n+1)/(2s)$ couplings.

D. Records and drive in one acceptance condition

Proposition 14 (Joint record–drive condition)**.** *If $chs\nu > 0$, the stationary loss satisfies $L_\infty(q) \leq L_*$ exactly when*

$$q \leq 1 - \frac{(1-2L_*)(1-c+chs)}{chs\nu}. \tag{32}$$

If the right-hand side is negative, no amount of calibration achieves L_* with that actuator.

Proof. Rearrange Eq. (27) using $1 - \Lambda = 1 - c + chs$. □

Take $L_* = 0.2$. A coherent drive with $N = 1$ fails even with a known interface, whose loss is 0.220330. For $N = 4$, 16 and 64 the minimal numbers of passive blocks are 57, 36 and 33; since q_n is nonincreasing, checking that $n-1$ blocks fail and n succeed establishes the minimum. With 32 blocks fixed, Eq. (32) requires $s \geq 0.9943781$, that is $\epsilon \leq 0.0056219$. Coherent half-turn drives then need $N \geq 109.10$, classical drives need $N \geq 98.93$ by Proposition 10, and by Theorem 8 every single-mode drive acting as a flip channel needs $N \geq 9.390$ (and $N + \lambda^2 \geq 9.503$). The Gaussian half-turn family reaches 9.957 and the computed grid variational minimum is 9.945. These minima refer to the declared block protocol and feedback rule, not to every possible experiment.

Corollary 15 (Scaling of the persistence cost)**.** *Among drives acting as flip channels, holding the stationary loss within δ of its exact-flip value at a known interface requires a drive occupation per activation that grows as $\delta^{-1/2}$ for optimal drives and as δ^{-1} for classical drives.*

Proof. At $q = 0$, L_∞ is a smooth decreasing function of s with nonzero derivative at $s = 1$, so δ is proportional to ϵ to first order. Theorem 8 and Proposition 9 give occupation of order $\epsilon^{-1/2}$ for optimal drives, and Proposition 10 together with the coherent family gives order ϵ^{-1} for classical drives. □

E. Resource accounting

The resources of the task are kept as a vector. Each task opportunity uses one reading, one probe and CNOT, a gate coin, a bath interval and one incoming drive of occupation N, prepared whether or not it is used. Each actual coupling adds switching work $\lambda^2\bar{\omega}$ at rate p. For $N = 16$ and 32 blocks, calibration uses 64 readings and 33 randomization opportunities carrying incoming energy $528\bar{\omega}$, with an expected 17.14 couplings and switching work $0.6606\bar{\omega}$. Over a task window of 60 steps the expected number of couplings is 6.494. The occupations 61.07 and 7.33 at $\epsilon = 0.01$ are input levels, and 7.49 minimizes input plus one activation's work; none of these is the total energy of an experiment, which would also include reading, preparation, records and the switching apparatus.

F. Finite-energy response readout

An outgoing stress can change a boundary trajectory, which can in turn change signal delays. Consider the particular readout in which these delays are the only branch-dependent signal. Use one calibrated proper clock, nonnegative probe Hamiltonians H_j on separate ports, and a joint probe–ancilla state independent of the unknown branch. The ancilla may be correlated with the probes but is not acted on by the signal. In branch $k = 0, 1$ the probes undergo $U_k = \exp[-i\sum_j H_j\tau_{kj}/\hbar]$. Preparation and final joint measurement can be arbitrary. The charged probe input is $E = \sum_j \operatorname{Tr}(\rho H_j)$, above the declared zero levels.

Theorem 16 (Sharp signed-delay energy bound)**.** *Let* $\Delta\tau = \max_j |\tau_{1j} - \tau_{0j}| > 0$ *and define*

$$\zeta = \max_{x>0} \frac{1-\cos x}{x} = 0.724611\ldots, \qquad z = \frac{\zeta E \Delta\tau}{\hbar}. \tag{33}$$

The trace distance of the two output states, including the retained ancilla, is at most $\sqrt{z(2-z)}$ *for* $0 \le z \le 1$*, and at most one otherwise. For equal priors, error at most* $\epsilon < 1/2$ *therefore requires*

$$E \ge \frac{\hbar}{\zeta\Delta\tau}\left[1 - 2\sqrt{\epsilon(1-\epsilon)}\right]. \tag{34}$$

This lower bound is sharp across the permitted probe families, including oppositely signed delay differences. It need not be sharp for a specified Hamiltonian or a readout with all delay differences of the same sign. If $\Delta\tau = 0$*, no such protocol distinguishes the branches.*

Proof. The scalar maximum occurs at the unique $x_* \in (0, \pi)$ satisfying $\tan(x_*/2) = x_*$; for $x \ge \pi$, $(1-\cos x)/x \le 2/\pi$, whereas the value at $2\pi/3$ gives $\zeta \ge 9/(4\pi) > 2/\pi$. Thus $1 - \cos x \le \zeta|x|$ for every real x. Purify the probe–ancilla input. The overlap of the two resulting pure states is

$$A = \operatorname{Tr}\rho\exp(-iK), \qquad K = \sum_j H_j(\tau_{1j} - \tau_{0j})/\hbar.$$

The Hamiltonians commute strongly, and their positivity gives $|K| \le (\Delta\tau/\hbar)\sum_j H_j$ by joint spectral calculus. Consequently $\operatorname{Re} A \ge 1 - z$. For $z \le 1$, the pure-state trace distance is $\sqrt{1-|A|^2} \le \sqrt{z(2-z)}$; discarding the purifying reference cannot increase it. Equal-prior error is at least $(1-D)/2$ for trace distance D. Solving this inequality gives Eq. (34).

For saturation, take two probes with equal excited energy $e = x_*\hbar/\Delta\tau$, and delay differences $+\Delta\tau, -\Delta\tau$. On the three states $|00\rangle, |e0\rangle, |0e\rangle$ prepare

$$|\psi\rangle = \sqrt{1-p}\,|00\rangle + \sqrt{p/2}\,(|e0\rangle + |0e\rangle), \qquad p = \frac{1 - 2\sqrt{\epsilon(1-\epsilon)}}{1-\cos x_*}.$$

Here $0 \le p < 1$, $E = pe$, and the branch overlap is exactly $1 - p(1-\cos x_*) = 2\sqrt{\epsilon(1-\epsilon)}$. The optimal binary measurement attains error ϵ and Eq. (34). □

Mean-energy quantum speed limits constrain distinguishable evolution, including arbitrary-fidelity targets [34, 35]. Here the relevant resource is the sum of the physical probe energies, not the signed generator's ground-state shift; this accounts for opposite delay differences without assigning a negative probe cost. A readable branch marker, changed output frequency or amplitude, branch-dependent preparation, or a signal interaction other than translation

falls outside the theorem. For example, an unpriced orthogonal marker would distinguish the branches at zero probe energy. Receiver, clock and preparation costs remain additional resources.

In HAFF-D, three echoes determine the coefficients of a supplied JT solution. Its source-compensated inverse permits known stresses in both null directions and propagates timing and source-calibration errors to the next coupled projection [29]. This connects a selection-dependent source, its response and a subsequent operation, conditional on the declared instrument. For its compact-pulse example the largest proper-delay difference is 0.05001338; a translation-only instrument with error 0.1 requires input energy at least $11.0374\hbar$ in that clock's units. The geometric inverse does not construct such a probe or its reflecting boundary. Its supplied bulk source-compensation test is not an energy–resolution demonstration. A joint apparatus must meet the readout bound, the actuator bound and its actual stress and entropy accounts without counting the same outgoing energy twice.

G. Finite-rate entropy export

Drive occupation and exported heat obey different bounds. If the maintained subsystem and its finite reused controller remove a net dimensionless entropy current $\dot{s} > 0$, and all of it leaves through a chiral conformal exhaust of capacity c_{ex} above a thermal input at temperature T, HAFF-D gives [29]

$$\dot{Q} \geq k_B T \dot{s} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} \dot{s}^2.$$

The current must include retained side information and controller consumption; the number of attempted flips alone does not determine it. The bound assumes an extensive long-window entropy-current account, zero chemical potential or the matched conserved-charge account, and one common clock. It is not an exact bound for every short packet. It remains positive at $T = 0$ for a thermal input, where ordinary Landauer heat is zero. For independent parallel exhausts at one temperature, c_{ex} is their total capacity.

A nonthermal drive changes this comparison. If its incoming energy and dimensionless entropy currents are e_{in} and σ_{in}, the same Gibbs argument gives

$$\dot{Q} + e_{\mathrm{in}} - E_{\mathrm{th}}(\sigma_{\mathrm{in}}) \geq k_B T_\sigma \dot{s} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} \dot{s}^2, \qquad E_{\mathrm{th}}(\sigma) = \frac{3\hbar\sigma^2}{\pi c_{\mathrm{ex}}}, \quad T_\sigma = \frac{6\hbar\sigma_{\mathrm{in}}}{\pi c_{\mathrm{ex}} k_B}.$$

The incoming energy above its entropy-matched thermal value can pay for export. In particular, a coherent beam with zero entropy is not a zero-energy reservoir. Its drive occupation and the additional heat must be accounted for separately.

This heat is neither the incoming occupation $N\bar{\omega}$ nor an extra charge to add to an outgoing energy already counted. It is a constraint on that energy when it carries the removed entropy. The scheduled drive calculation above does not establish its exhaust-capacity assumptions. A complete apparatus must price reading, recovery, sensing and entropy transport in the same cycle.

VIII. CALIBRATING AN INTERFACE WHILE IT CARRIES QUANTUM CONTENT

Preparing test states before storage is not the only way to learn an interface. The outgoing field can carry information about the coupling without carrying the logical content that the apparatus is trying to keep. We give the instrument, its loss under incomplete collection, and a coherent implementation.

On the two-qubit code $\mathcal{C} = \mathrm{span}\{|01\rangle, |10\rangle\}$, let Z_L have eigenvalues $+1, -1$. The coupling $Z_1 + \alpha a Z_2$ restricts to mZ_L, $m = 1 - \alpha a$, with unknown $\alpha > 0$ and chosen command a. Write

$$Q_f = \sqrt{k}(b + b^\dagger), \qquad U_m = e^{-imZ_L Q_f}, \qquad k > 0, \quad \Pi = (-1)^{b^\dagger b}.$$

The independently prepared incoming mode is vacuum. A beam splitter collects fraction ξ of this mode, leaving fraction $1 - \xi$ inaccessible.

Proposition 17 (A content-preserving calibration instrument)**.** *Measure the collected mode's parity and apply Z_L for an odd outcome. The corrected instrument, including an arbitrary retained reference, is*

$$\mathcal{I}_\pm(\rho) = p_\pm(a)\mathcal{D}_{\xi,m}(\rho), \qquad p_\pm(a) = \frac{1 \pm e^{-2k\xi m^2}}{2}, \tag{35}$$

where $\mathcal{D}_{\xi,m}$ fixes the populations and multiplies logical coherence by $e^{-2k(1-\xi)m^2}$. At full collection each outcome is a scalar multiple of the identity channel. Fresh-port queries chosen from earlier records can therefore calibrate the interface without replacing its held state.

Proof. The accessible displacement has the expansion $I_L \cos(m\sqrt{\xi}Q_A) - iZ_L \sin(m\sqrt{\xi}Q_A)$. On the vacuum the two field vectors have opposite parity, with squared norms p_+, p_-. The conditional pulse makes both logical factors I_L. The independent inaccessible displacement commutes with this correction and leaves $\mathcal{D}_{\xi,m}$. Each branch is thus Eq. (35), not merely an equality of its unconditional channel. Composition proves the adaptive statement; the same computation tensors with any reference. □

Environment-assisted correction of unitary Kraus branches is a known general mechanism [3, Proposition 2]. Here its records are also the data for learning the same interaction's gain and collection fraction. Repeated ancilla-based parity measurements have been implemented for a cavity mode [5]; collecting the full mode of a travelling chiral field is an additional requirement, not supplied by that experiment.

Partial collection prices the calibration itself. For N queries at each of $a = \pm h$, with fixed α, ξ, the held coherence is multiplied by

$$\exp[-4Nk(1-\xi)(1+\alpha^2 h^2)]. \tag{36}$$

This loss occurs before any subsequent operating benefit. With $k = .2$, $\alpha = 1.1$, $h = .5$ and $N = 4096$, keeping the calibration's dephasing error below .01 needs $1 - \xi \le 4.7335 \times 10^{-6}$. At $\xi = .99$ its error is essentially $1/2$. More records are not automatically better for held content.

The collection fraction in the preceding instrument is realized by the resonant and finite-band protocols of Appendix A. Their finite-packet, waveform and renewal estimates retain the assumptions needed for actual access. The following echo concerns the resulting available mode.

Proposition 18 (Coherent parity echo)**.** *Suppose the relevant mode remains available for joint control. The bounded involution*

$$R = |0_L\rangle\langle 0_L| \otimes I + |1_L\rangle\langle 1_L| \otimes \Pi = I_L \otimes P_{\text{even}} + Z_L \otimes P_{\text{odd}}$$

satisfies, on the quadrature domain,

$$R(Z_L \otimes Q_f)R = I_L \otimes Q_f, \qquad RU_m R = I_L \otimes e^{-imQ_f}. \tag{37}$$

This protects the body's marginal for every mode input and even for an initially correlated body–mode state. For an independent vacuum input, the first R acts trivially; the final R alone disentangles the body and mode.

Proof. $\Pi Q_f \Pi = -Q_f$. In the $+1$ logical block the conjugated generator is Q_f, and in the -1 block it is $-\Pi Q_f \Pi = Q_f$. Functional calculus gives the unitary identity. A unitary on the mode alone cannot change the body's reduced state. On vacuum $\Pi|0\rangle = |0\rangle$. □

One qubit meter implements R coherently. Prepare it in $|0\rangle_C$, apply

$$M = \exp[-i(\pi/2)P_{\text{odd}} \otimes Y_C], \qquad C_Z = I_L \otimes |0\rangle\langle 0|_C + Z_L \otimes |1\rangle\langle 1|_C,$$

and then apply $M^\dagger C_Z M$, with the tensor identities understood. The meter computes parity, controls the body phase and uncomputes parity; it returns exactly to $|0\rangle_C$. Bounded Hamiltonians $g(t)P_{\text{odd}} \otimes Y_C$ and $\gamma(t)(I_L - Z_L)/2 \otimes |1\rangle\langle 1|_C$ suffice, with pulse areas $\pi/2$ and π. Amplitude bounds impose corresponding nonzero durations. No classical syndrome bit has to be erased in this coherent correction. Producing a retained classical calibration record is a separate operation. Pulse area is not a heat bill: the control field, capture, clock and support still need their physical energy account. Half-diamond errors of supplied primitive channels accumulate at most by their sum, by telescoping compositions; such a bounded-channel error alone does not bound an unbounded stress observable.

A retained oscillator offers a direct alternative to the spectral meter. In its interaction picture, with $n = b^\dagger b$ and $P_1 = |1_L\rangle\langle 1_L|$, the dispersive Hamiltonian $H_R(t) = \hbar\chi(t)P_1 \otimes n$ implements R when $\int \chi(t)dt = \pi$. Such number-controlled parity gates are established cavity operations [5, pp. 2–3]. For a degenerate logical body, $\chi \ge 0$ and oscillator energy $\hbar\omega n$, the total Hamiltonian is bounded below. If χ starts and ends at zero and the gate is isolated, $P_1 n$ is conserved, so its net prescribed-switch work is exactly zero:

$$W_R = \hbar \int \dot{\chi}\, \langle P_1 n\rangle dt = 0.$$

The oscillator's stored energy, gate duration and controller are not zero. The integrated expected interaction energy is $\pi\hbar\langle P_1 n\rangle$; $\chi \le \chi_{\max}$ gives duration at least $\pi/\chi_{\max}$. If the pulse area has error δ, its half-trace error, with any reference and an input occupation bound $\langle n\rangle \le \bar{n}$, is at most $\min\{1, \sqrt{2|\delta|\bar{n}}\}$. Indeed the relative unitary is $e^{-i\delta P_1 n}$, and the purified vector error squared is bounded by $2|\delta|\langle P_1 n\rangle$, using $1-\cos x \le |x|$. This is an energy-constrained error, not an unrestricted diamond bound. It gives an explicit retained-oscillator gate account; it does not price trapping a broadband travelling mode or operating the classical switch.

A. Thermal inquiry with a binary record and a finite bank

The echo gives an actual thermal inquiry protocol on the retained oscillator. Let its input be the centered Gibbs state τ_T of $H_b = \hbar\omega n$, with $n_T = (e^{\hbar\omega/(k_B T)} - 1)^{-1}$ and $\nu_T = 2n_T + 1$. This input is independent of the held body, its old reference and earlier records. Both parity gates in $RU_m R$ are used: the first is generally consequential on a thermal input. The result is $\rho_{L\mathrm{old}} \otimes D_m \tau_T D_m^\dagger$, where $D_m = e^{-im\sqrt{k}Q}$. Thus the body and its entire old reference are preserved, while the oscillator has

$$\langle P\rangle = -2m\sqrt{k}, \qquad \mathrm{Var}(P) = \nu_T, \qquad E_b - E_{b,\mathrm{in}} = \hbar\omega k m^2.$$

The displaced Gibbs state has the same entropy as τ_T.

A quadrature mean is not yet a saved record. Prepare a fresh degenerate qubit C in $\eta_v = (I + vZ_C)/2$, $0 < v \le 1$, independently of the oscillator and all earlier outputs. Couple it through

$$V_\kappa = e^{-i\kappa P \otimes Y_C}, \qquad \kappa > 0,$$

then measure X_C and store the outcome $Y = \pm 1$. The preparation, orientation, measurement and storage are declared operations.

Proposition 19 (Thermal binary inquiry)**.** *For this protocol the instrument on the held body and its old reference is*

$$\mathcal{J}_y(\rho_{L\mathrm{old}}) = \frac{1 + y\mu_T(m)}{2}\rho_{L\mathrm{old}}, \qquad \mu_T(m) = -ve^{-2\kappa^2\nu_T}\sin(4\kappa m\sqrt{k}). \tag{38}$$

The read interaction adds oscillator energy $\hbar\omega\kappa^2$ before conditioning on the record. If $m = \pm m_0$ with known $m_0 > 0$, $0 < \kappa \le \pi/(8m_0\sqrt{k})$ gives sign gain $\gamma_T = ve^{-2\kappa^2\nu_T}\sin(4\kappa m_0\sqrt{k}) > 0$. For J fresh trials, the sign estimate from $-\sum_j Y_j$ has error at most $e^{-J\gamma_T^2/2}$, including ties as errors.

Proof. The echo acts only on the independent oscillator, so all subsequent oscillator–port operations factor from $\rho_{L\mathrm{old}}$. Conjugation gives $V_\kappa^\dagger X_C V_\kappa = \cos(2\kappa P)X_C + \sin(2\kappa P)Z_C$. The Gibbs characteristic function is $\langle e^{itP}\rangle = e^{-t^2\nu_T/2}$, and displacement changes it by $e^{-2itm\sqrt{k}}$. Taking the qubit expectation proves Eq. (38). Also $V_\kappa^\dagger Q V_\kappa = Q + 2\kappa Y_C$, while P is fixed. Since the independent port has $\langle Y_C\rangle = 0$ and $Y_C^2 = I$, $n = (Q^2 + P^2 - 2)/4$ increases in mean by κ^2. Each sign-aligned binary variable has mean γ_T; Hoeffding's inequality gives the stated bound. Independence here follows from fresh inputs and the renewal below, rather than from reusing a disturbed mode. □

This record reads a sign that parity cannot resolve. Its orientation reference is an additional permission. The sine is periodic, so one setting does not identify arbitrary continuous m; an unknown initial quadrature mean also confounds it. If the mode initially shares correlations with the body, the echo still preserves the unconditional body state, but measuring the mode may steer it. That case does not give Eq. (38).

The pulse can have bounded amplitude and finite duration. In the laboratory frame choose a smooth real $g(t)$ of compact time support, with $\int g(t)dt = \kappa$, and set

$$H_{\mathrm{read}}(t) = \hbar\omega n + \hbar g(t)[P\cos(\omega t) + Q\sin(\omega t)] \otimes Y_C. \tag{39}$$

Its interaction-picture generator is exactly $\hbar g(t) P \otimes Y_C$. The final laboratory evolution includes a known free rotation; subsequent pulses and read axes use the same clock. Completing the square gives $H_{\mathrm{read}}(t) \ge -\hbar g(t)^2/\omega$. Adding the scalar $\hbar g(t)^2/\omega$ makes it nonnegative and changes only an overall phase; the scalar vanishes at both endpoints. For $|g| \le g_{\max}$ the duration is at least $\kappa/g_{\max}$. The net prescribed-switch work is the endpoint free-Hamiltonian energy increase $\hbar\omega\kappa^2$ in this isolated, degenerate-port model.

The echo's displacement admits the same finite pulse construction, with laboratory force $\hbar m\sqrt{k} f(t)[Q\cos(\omega t) - P\sin(\omega t)]Z_L$ and $\int f(t)dt = 1$. Completing the square bounds its Hamiltonian below by $-\hbar k m^2 f(t)^2/\omega$; a finite command range keeps this bound uniform. The source pulse supplies $\hbar\omega k m^2$ on the Gibbs input. The two parity gates have the finite dispersive realization above. These rotating profiles supply phase and clock access explicitly; their implementation by an autonomous controller is a further physical question.

Proposition 20 (Finite thermal renewal with retained outputs)**.** *Take a bank of J independent same-frequency oscillators in τ_T, independent mixed read ports, and storage for J binary records. After each echo and read, couple b to the next fresh bank oscillator through $\hbar h(t)(b^\dagger c + c^\dagger b)$ with area $\pi/2$. The active oscillator returns exactly to τ_T, independent of all earlier retained outputs. Its entire post-read state and correlations are transferred to the spent bank oscillator, up to known phase rotations. After summing over the new outcome, the entire old body/reference/history state is preserved, including for commands selected from earlier records. If $|h| \le h_{\max} < \omega$, the total contact Hamiltonian is bounded below and its duration is at least $\pi/(2h_{\max})$. A pulse off at both endpoints has zero net prescribed-switch work. For fixed m and κ, the spent oscillator's mean excess energy above its incoming Gibbs value is*

$$\Delta E_{\text{spent}} = \hbar\omega(km^2 + \kappa^2). \tag{40}$$

This is a finite J-trial realization of Eq. (38)*.*

Proof. The contact sends $b \mapsto -ic$, $c \mapsto -ib$. For any joint post-read state ρ_{bA}, where A contains the held body, old reference, port and records, the output is $\tau_{T,b} \otimes \rho'_{cA}$. Gibbs states are invariant under the known phase rotation. This identity transfers correlations instead of tracing them out. The two instantaneous normal frequencies are $\omega \pm h$. Total occupation and $b^\dagger c + c^\dagger b$ are conserved during the contact, which proves the stability and zero switch-work statements as in Proposition 21. The preceding two displacement accounts prove Eq. (40) after averaging over the retained record. For an old classical history r, the new record appends the kernel $p(y \mid m_r)$ to its existing conditional body/reference state. Summing over y leaves each old block, and hence their full joint state, unchanged. Induction proves renewal and this nonselective preservation for adaptive commands. Conditioning only on a new outcome can reweight old history blocks and change the reader's body/reference state; it need not preserve that conditional state. For a fixed unknown sign and fixed settings the new inputs give independent trials. □

For example, $k = .2$, $m_0 = .7$, $v = .6$ and $\kappa = \min\{\pi/(8m_0\sqrt{k}), 1/(2\sqrt{\nu_T})\}$ give sufficient trial counts 178, 247 and 593 for error .05 at $n_T = .2$, .5 and 2. The respective excess oscillator energies per trial, in units of $\hbar\omega$, are .276571, .223 and .148. Higher temperature weakens the record even though this choice reduces the read-pulse energy. A bank consumes one fresh Gibbs oscillator and one mixed read port per trial, retains the spent oscillator and a binary record, and pays both parity gates, the source/read profiles and the resonant contact duration. The initial active Gibbs oscillator is needed in addition to the J fresh bank inputs. Preparation, transport, readout, storage, orientation and the clock/support must be charged in addition.

Neither term in Eq. (40) is an entropy-export certificate. The source displacement preserves entropy; the read operation changes correlations with its port and record. Replenishing the bank or erasing spent outputs requires their joint state, including side information. This construction supplies repeated thermal inquiry for each finite bank. It does not supply the chiral capacity of Section VII, an infinite thermal line or the total covariant stress of a controller.

For a centered independent input the original outgoing source is $m^2 F_f$, $F_f = (f')^2/4$. The coherent echo leaves this coupling-added stress in the field while preserving content. Its implementation requires an available mode and controlled parity; it supplies neither a local capture apparatus for a whole continuous field nor a sustained thermal exhaust. The finite oscillator identity and the moving-field apparatus are different levels of the construction.

IX. RELATIONS TO OTHER WORK

A. Q-RAIF Papers A and C

Paper A [30] looks for the relative sign in translation-invariant local unitary propagation. When all bands share one quadratic shell, the propagation symbol obeys Euclidean Clifford relations. Its form $\eta = \omega^2 - Q(q)$ has one positive frequency direction and d negative spatial directions; the mass term together with the spatial generators yields $Cl(1, d)$. Reversing the overall metric convention reverses these signs, not their relative opposition. By Sylvester's law neither positive G nor K of Section V can be brought to that indefinite form.

Section V relates the spatial shell to measured work under explicit alignment, while keeping their domains and the indefinite spacetime form distinct. Paper C [31] asks what a coupling supplies in place of an assumed representation map. For the coupling XQ_g of this paper the algebra represented exactly is the commutant of X, for every drive state, and the defect on the remaining controls reproduces the energy–accuracy bound of Theorem 8.

The mass–energy paper [32] now uses the same quadrature interaction to realize a quantum-walk coin. With a fresh Gaussian drive each step, its mean angle fixes the rest resonance, while drive fluctuations set its decay rate. At input occupation N the smallest Gaussian rate is $\theta^2/[2\tau N(N+1)]$, attained by a displaced squeezed state. This connects the price of control to the persistence of a spectral readout. The rate is an open-channel linewidth, not a new spectral mass, and shift, preparation and port reset costs are separate.

B. Quantum control theory

Lie-algebraic control characterizes reachability through the algebra generated by drift and control Hamiltonians [14]. Stabilization against perturbations also needs magnitudes and costs. In the realized actuator these come from the physical coupling: the rotation angle from the drive amplitude, the error from the drive's quadrature fluctuations, and the cost from the field stress. A Clifford quadratic form is not needed to supply them.

C. Decoherence-free subspaces

Decoherence-free subspaces [19, 20] are protected passively by symmetry. The feedback task gives the active counterpart. The held-code instrument, coherent echo and finite collection protocols also join passive encoding to priced inquiry and correction: their exact identities protect a supplied code, while collection loss and gate error determine what survives. Brain Anchor [4] uses records of this interface for finite-window inquiry and maintenance decisions. Forming the code and its physical permission from a prior history is a further mechanism, not a consequence of encoding it.

D. The Imprint Chain and HAFF

The Imprint Chain [36] supplies the instrument, the calibration and the task, with an exact supplied flip. Replacing that flip by the field coupling raises the calibrated loss from 0.199631 to 0.202127 at $N = 16$ and makes the drive energy part of the feasibility condition. Each coupling also adds the stress F_g to the outgoing field. The HAFF gravity paper [29] uses this outgoing field as a matter source in a Jackiw–Teitelboim model; there, calibration changes the mean source through the coupling rate p of Eq. (28). Selecting the coupled projection changes its represented algebra and source even at fixed activation schedule. A boundary readout can then update the next projection, with known probe stresses and bounded errors included in the inverse. Section VII F supplies one energy constraint on a specified realization of that return.

E. Quantized control fields

The effect of field quantization on gate accuracy has a substantial literature. For Jaynes–Cummings-type couplings, conservation of excitation number leads to error bounds controlled by photon-number fluctuations, of order $1/\bar{n}$ for coherent fields [24]; coherent-field control limits for such models were analysed in Ref. [25], and simultaneous driving of several atoms by one quantized pulse in Ref. [26]. The coupling studied here is a quadrature coupling of a degenerate bit and conserves no excitation number, which is why squeezing changes the scaling from ϵ^{-1} to $\epsilon^{-1/2}$. Theorem 8 gives the quadrature-specific bound and explicit near-attainment under these assumptions. A covariant stress tensor for a finite detector, including its localization, is constructed in Ref. [23]; it shows why the apparatus stress cannot simply be omitted.

X. DISCUSSION

For gradient-type feedback the algebra must compose associatively and the descent metric must be positive. When control is realized by a field, the structure that limits persistence is the canonical commutation relation of the drive quadratures: $[Q, P] = 2i$ fixes the trade-off between flip accuracy and drive occupation in Theorem 8, and it is the reason squeezed drives do better than classical ones. For the fixed Pauli flip the commutation relation of the driving mode sets the accuracy cost. For a family of control operators, Proposition 6 additionally identifies when their quadratic structure fixes a state-independent centered-input work form. Persistence has a closed-loop proof in this model, with an explicit Lyapunov function, and the drive energy and calibration records enter one feasibility condition.

Several inputs remain supplied. The free chiral field, the degenerate bit, the clock and switch, the drive ports and the instrument family are given. The feedback task also supplies the success probability for compensated randomization and scheduled beams. Its analysis concerns Z-diagonal bit states and flip channels; complex $\chi_\sigma(2\lambda)$ instead rotates Z into Y. The held-content construction concerns arbitrary logical inputs. Its finite-band realization prepares and collects a specified packet, with a finite waveform, clock tolerances and independent vacuum exhausts. It returns a

coherent meter; the carrier, detector, port selection and vacuum supply remain separate inputs. The stress of the switch and its support is not computed; outgoing-field stress is used outside the apparatus.

The thermal bank identifies what renewal has to preserve. A fresh input restores the oscillator only because the previous state, including its record correlations, moves to a spent output. Bank replenishment and memory erasure must therefore be priced on that joint output, as in the information thermodynamic account [9, 10]. The chiral capacity bound constrains a specified entropy flow; the finite oscillator construction gives an implementable inquiry and transfer, rather than that flow's attainability.

Repeated propagation in one field remains a stronger problem. Proposition 12 proves that shared drives can preserve each marginal channel while correlating failures. A persistence calculation on that joint history may therefore have a different energy scaling from the fresh-drive task. The other unresolved connection is formation: which interactions and retained records produce the code, access permissions and controller used here? T-DOME, Imprint and Brain Anchor address that mechanism; the present results constrain the resources of the operations it must supply.

XI. CONCLUSION

The connection between control and propagation becomes precise only after a physical condition is specified. In the centered-field calibration studied here, a linear Hermitian control family has state-independent switching work exactly when it has a scalar quadratic square. A calibrated use of those same generators for propagation then gives a positive pullback relation between work and the spatial shell. The time direction and mass anticommutator enter separately. Persistence alone supplies none of these identifications.

The same field coupling gives a finite actuator whose stress, accuracy and drive energy can be calculated together. Squeezed inputs approach the $\epsilon^{-1/2}$ occupation bound, while classical drives require order ϵ^{-1}. Calibration and physical actuation enter a single feedback feasibility condition. Content-preserving inquiry, finite port renewal and the separate delay-readout bound extend the accounting to both directions of an interface. The remaining formation problem is which physical histories produce the aligned couplings, permissions and controller, rather than merely supplying them. The conditional bridge locates that question at the coupling itself; it does not replace it with the existence of a well-priced device.

Appendix A: Physical collection of the inquiry mode

These constructions support the mode access used in Section VIII. They distinguish a resonant stored mode, a finite packet and a finite-band travelling input; none substitutes a free capture or renewal operation for a physical interaction.

1. A resonant collection port

The collection fraction can itself be realized by a finite-duration interaction. This construction starts with a retained source oscillator b and a receiver oscillator c of the same frequency ω, with c initially in vacuum. It does not assume that a broadband travelling mode has already been trapped by this interaction. In the interaction picture put

$$H_{\mathrm{cap}}(t) = \hbar g(t) C, \quad C = b^\dagger c + c^\dagger b, \quad \theta = \int g(t)\,dt, \quad U_{\mathrm{cap}} = e^{-i\theta C}.$$

Proposition 21 (Finite resonant collection)**.** *For $0 \le \theta \le \pi/2$, the receiver collects $\xi = \sin^2\theta$ of the source mode. Corrected receiver-parity readout realizes Eq. (35) with this ξ. If $|g| \le g_{\max} < \omega$, the total Hamiltonian $\hbar\omega(n_b + n_c) + H_{\mathrm{cap}}$ is bounded below and the pulse needs duration at least $\theta/g_{\max}$. A pulse starting and ending at zero does no net prescribed-switch work in this isolated two-mode model. For the independently prepared vacuum encounter, the mean occupations after collection are ξkm^2 and $(1-\xi)km^2$. Their sum is km^2.*

A pulse-area error δ changes ξ by at most $|\delta|$. For any input with $\langle n_b + n_c\rangle \le \bar{n}$, including a retained reference, its half-trace error is at most $\min\{1, \sqrt{2|\delta|\bar{n}}\}$.

Proof. The Heisenberg transforms are $b \mapsto b\cos\theta - ic\sin\theta$ and $c \mapsto c\cos\theta - ib\sin\theta$. The conditional coherent states produced by U_m therefore split into receiver and remainder amplitudes with squared fractions $\xi, 1-\xi$. The receiver's known phase does not change parity. Applying Proposition 17 proves the instrument and the occupation account, for any logical input.

The normal modes $(b \pm c)/\sqrt{2}$ have instantaneous frequencies $\omega \pm g$, which proves the lower bound. The amplitude bound proves the duration statement. Both C and total occupation commute with the Hamiltonian, so $\int \langle \partial_t H_{\mathrm{cap}} \rangle dt = \hbar \langle C \rangle [g(t_{\mathrm{end}}) - g(t_{\mathrm{start}})] = 0$. Finally $|\sin^2(\theta + \delta) - \sin^2 \theta| \le |\delta|$ and $|C| \le n_b + n_c$. The purified-vector estimate $2\langle 1 - \cos(\delta C) \rangle \le 2|\delta|\bar{n}$ proves the energy-constrained error bound. □

This account includes transferred energy and a gate clock. Receiver parity control, reading and retaining a syndrome, preparing the next empty receiver, and supplying the switching apparatus remain separate resources. Emptying the receiver exports energy; it is not included in the zero net switch work above. Restoring a degraded collection setting also needs a stated physical permission and price. Brain Anchor [4] uses such a permission in a finite model where the same content survives inquiry, operation and maintenance. Resonant port capture, local capture of a continuous field and sustained thermal export are three different implementation questions.

Whole-mode control can be reduced to successive port contacts in a finite packet model. Take n source oscillators of frequency ω and one vacuum receiver c. For real $f_j \ge 0$, $\sum_j f_j^2 = 1$, form the packet mode $b_f = \sum_j f_j b_j$. Sequential source interactions $e^{-im\sqrt{k} f_j Z_L (b_j + b_j^\dagger)}$ produce the same packet displacement as $e^{-im\sqrt{k} Z_L (b_f + b_f^\dagger)}$. After contacting port j, couple only that port to c through the resonant Hamiltonian above, with area $0 \le \theta_j \le \theta_{\max} \le \pi/2$. Released ports are inaccessible. The receiver's packet amplitude, up to its known common phase, obeys

$$y_0 = 0, \qquad y_j = y_{j-1} \cos \theta_j + f_j \sin \theta_j.$$

Proposition 22 (Sequential local-port collection)**.** *The receiver collects fraction $\xi = y_n^2$ of the packet, and its corrected parity instrument is Eq.* (35)*. Among these permitted real resonant contacts, final collection is maximized by*

$$\theta_j = \min\{\theta_{\max}, \arctan(f_j / y_{j-1})\},$$

with the first angle $\pi/2$ before applying the cap when $y_{j-1} = 0 < f_j$. When $y_{j-1} = f_j = 0$, choose $\theta_j = 0$. For a uniform packet and $\theta_{\max} = .5$, the first five contacts are capped and the remaining ones are unconstrained. For $n \ge 5$ the exact optimum is

$$\xi = 1 - \frac{d}{n}, \qquad d = 5 - \cot^2(.25)(1 - \cos^5(.5))^2 = 1.47392688\ldots.$$

Thus 148 uniform ports suffice for collection at least .99, and fewer do not under the same cap. With no cap below $\pi/2$, the angles $\theta_1 = \pi/2$, $\theta_j = \arctan(1/\sqrt{j-1})$ collect a uniform finite packet completely.

Proof. Passive mode mixing sends the packet's conditional coherent displacement to receiver amplitude $-iy_n$ and a remainder of squared norm $1 - y_n^2$. Orthogonal input modes are vacuum. Their overlaps prove the instrument. Each step maximizes $y \cos\theta + f_j \sin\theta$ at the stated angle. The maximum is nondecreasing in $y \ge 0$, because all permitted cosines are nonnegative. Induction therefore makes this stepwise optimum a final optimum. For a uniform packet, the first capped steps give $y_j = n^{-1/2} \cot(.25)(1 - \cos^j(.5))$. After the fifth, the interior optimum has $y_j^2 = y_{j-1}^2 + 1/n$. This proves the formula and threshold. The uncapped recursion gives $y_j = \sqrt{j/n}$. □

Each contact has a Hamiltonian bounded below when $g_{\max} < \omega$, needs duration at least $\theta_j / g_{\max}$ and conserves total occupation. The source prepares mean total occupation km^2, the receiver holds ξkm^2 and the released ports hold $(1-\xi)km^2$. Parity control is needed only on the retained receiver. For contact-area errors δ_j, the half-trace error is at most $\min\{1, \sum_j \sqrt{2|\delta_j|\bar{n}}\}$ when total occupation is bounded in mean by $\bar{n}$; telescope the preceding energy-constrained bound. Source switching, receiver reuse and detector errors require their own account. These are local contacts in a finite collision architecture, not a proof that a specified relativistic wave packet admits that architecture or that its clocks and references can be sustained indefinitely.

2. Positive-frequency continuous collection

A specified continuous field can also supply the collection port. Let $\omega = \Omega + \delta$, $|\delta| \le B < \Omega$, and couple it to a retained oscillator c. With $[a_\delta, a_{\delta'}^\dagger] = \delta(\delta - \delta')$, take a real even form factor g, $J = g^2$ and $G = \|g\|_2$. Put

$$H(h) = \hbar\Omega n_c + \hbar \int_{-B}^{B} (\Omega + \delta) a_\delta^\dagger a_\delta d\delta + \hbar h(t) \int_{-B}^{B} g(\delta)(c^\dagger a_\delta + a_\delta^\dagger c) d\delta, \qquad |h| \le 1. \tag{A1}$$

This is an excitation-preserving effective Hamiltonian with a specified band-smeared coupling. Its frequency form factor gives spatial tails; it is not a strictly point-local relativistic detector. No negative physical frequencies or Markov

extension of the band are used. Matched pulse absorption by a tunable resonator is an established mechanism [2]; the question here is how its finite-band realization enters the held-content instrument and source account.

In the frame rotating at Ω, the one-particle operator is

$$A = \begin{pmatrix} 0 & \langle g| \\ |g\rangle & \delta \end{pmatrix}, \quad S = e^{-iAT}, \quad r(T) = \langle c|S|c\rangle.$$

The capture pulse has $h = 1$ on $[0, T]$ and is otherwise off. Deviations from this prescribed pulse are bounded below.

Proposition 23 (Matched finite-band capture)**.** *If $\Omega > B + G$, every Hamiltonian in Eq.* (A1) *is bounded below by $\hbar(\Omega - B - G)N$. Write $\xi_0 = 1 - |r(T)|^2 > 0$. Among normalized incoming field modes, the mode*

$$f = -P_{\text{field}} S^\dagger |c\rangle / \sqrt{\xi_0}$$

maximizes collection and is collected with fraction ξ_0. It can be prepared from independent vacuum by a finite prescribed body-controlled linear drive with envelope $u(s) = r(T - s)$, $0 \le s \le T$. For even J, r and this envelope are real. The prepared field amplitudes are

$$m\beta w_u(\delta), \qquad w_u(\delta) = -ig(\delta) \int_0^T u(s) e^{i\delta s} ds, \quad w_u = -P_{\text{field}} S^\dagger |c\rangle, \quad \beta = \sqrt{k/\xi_0}. \tag{A2}$$

For an arbitrary prepared envelope and $w_u \neq 0$, replace k by $k_{\text{eff}} = \beta^2 \|w_u\|^2$ and use $\xi = |\langle c|S(0, w_u)\rangle|^2 / \|w_u\|^2$. Corrected receiver-parity readout then realizes Eq. (35) *with these actual parameters.*

Proof. $\|A\| \le B + G$ and second quantization prove the lower bound. The norm of the field part of the receiver row is $\sqrt{\xi_0}$; Cauchy–Schwarz proves the optimum. Its orthogonal complement has zero collection, so the statement is not uniform over unknown packets.

Eliminating the field component of $S^\dagger|c\rangle$ gives Eq. (A2). Reflection $\delta \mapsto -\delta$, field sign reversal and complex conjugation define an antiunitary taking A to $-A$ and leaving its propagator invariant. Hence r is real. One explicit preparation places the band-smeared source at $x = -T$ and the receiver at $x = 0$, with propagation speed one. Drive the source during $t \in [-T, 0]$ with envelope $u(t+T)$ and a prescribed carrier phase Ω. Explicitly its interaction-picture Hamiltonian is

$$H_s^I(t) = \hbar m \beta Z_L u(t + T) \int_{-B}^{B} g(\delta)[a_\delta^\dagger e^{i\delta(t+T)} + a_\delta e^{-i\delta(t+T)}] d\delta.$$

Its displacement is $m\beta w_u$; the receiver is coupled only after this drive. The source Hamiltonian is linear in the band field and proportional to mZ_L. Its time-ordering commutator is a scalar times Z_L^2, so it changes only an overall phase. Completing the square bounds its instantaneous energy below since $\Omega - B > 0$ and $\int J/\omega$ is finite.

Passive field evolution splits the conditional coherent amplitudes into a receiver component of squared fraction ξ and a remainder of fraction $1 - \xi$. Their vacuum overlaps give the claimed instrument for arbitrary logical input and retained reference. Initial independence and parameters fixed within an encounter are required. □

The propagation prescription is within the band-smeared model: a strictly band-limited field operator has no compact spatial support. The construction supplies a drive and a receiver, not their formation from unknown interactions. For real envelopes and even g, the free input and output energies are

$$E_{\text{source}} = \hbar\Omega k_{\text{eff}} m^2, \qquad E_c = \hbar\Omega\xi k_{\text{eff}} m^2, \qquad E_{\text{free,out}} = \hbar\Omega(1 - \xi) k_{\text{eff}} m^2. \tag{A3}$$

The same reflection symmetry makes the mean detuning energy zero before and after collection. Consequently the net prescribed capture-switch work is zero in this family. The complete source drive supplies E_{source}. For other packets the capture work need not vanish: at mean occupation $\bar{n}$, its magnitude is at most $2\hbar B\bar{n}$. Detector, carrier, clock and support costs are separate from these accounts.

Proposition 24 (A finite waveform certificate)**.** *Let $J(\delta) \le J_0$. Approximate $r(T - s)$ by linear interpolation of node values on L uniform intervals, each rounded on a prescribed grid of spacing q. Include a further uniform envelope error η_u. With $\nu = \sqrt{\xi_0}$,*

$$\sigma = \sqrt{2\pi J_0 T} \left[\frac{G^2 (T/L)^2}{8} + \frac{q}{2} + \eta_u \right]$$

bounds the prepared field-vector error relative to Eq. (A2)*. An interval enclosure error for node values is added inside the brackets. Let duration and arrival errors be bounded by* d_T, d_t*, and the capture pulse have* $\int |h_{\mathrm{actual}} - h|dt \le d_h$*. Put* $\varepsilon = (B+G)d_T + Bd_t + Gd_h$*. If* $\sigma < \nu$*, then*

$$1 - \xi \le \left[\frac{|r|\nu + \sigma + \varepsilon(\nu+\sigma)}{\nu - \sigma}\right]^2, \qquad \left(1 - \frac{\sigma}{\nu}\right)^2 \le \frac{k_{\mathrm{eff}}}{k} \le \left(1 + \frac{\sigma}{\nu}\right)^2. \tag{A4}$$

Arrival and duration bounds here concern collection and the logical parity instrument, which are insensitive to the common carrier phase.

Proof. $|r''(s)| = |\langle Ac|e^{-iAs}|Ac\rangle| \le G^2$ gives the interpolation bound. Plancherel over the entire frequency line, followed by restriction to the band, gives $\|w_u - w_{\mathrm{ideal}}\| \le \sigma$. Duhamel bounds the carrier-referenced propagator error by ε. The ideal outgoing field remainder has norm $|r|\nu$. Adding preparation and propagator errors to this norm and dividing by the actual input norm, at least $\nu - \sigma$, proves the first bound. The second follows by the same input-norm bounds. □

For a concrete continuum certificate use

$$J(\delta) = \frac{1}{2\pi}\left[1 - (\delta/B)^2\right]^2, \quad \Omega = 20, \quad B = 5, \quad T = 16.$$

Frequencies are measured in units of $2\pi J_0$ and times in its reciprocal. Then $G^2 = 8/(3\pi)$. The exact even bath moments are $\mu_{2j} = 16J_0B^{2j+1}/[(2j+1)(2j+3)(2j+5)]$. If $M_n = \langle c|A^n|c\rangle$, then

$$M_0 = 1, \qquad M_{2n} = \sum_{j=0}^{n-1} \mu_{2j} M_{2(n-1-j)}, \qquad r(t) = \sum_{n=0}^{\infty} \frac{(-1)^n M_{2n} t^{2n}}{(2n)!}.$$

Decomposing each departure and return to c proves the recurrence. The degree-P Taylor remainder is at most $[(B+G)T]^{P+1}/(P+1)!$ by self-adjoint functional calculus. Directed intervals at 120 decimal digits and $P = 400$ give $r(16) = 0.000061743299929498335 73\ldots$ with remainder below 1.53×10^{-79} and $\xi_0 > 0.99999999618$.

Choose $L = 512$, $q = 1/65536$, $\eta_u = d_T = d_t = d_h = 10^{-5}$. The 513 rounded node values fit the grid $[-1/64, 1]$, whose 66561 labels need 17 bits each. The fixed table uses 8721 logical bits, apart from its decoder, interpolator, carrier and clock. A separately encoded normalization amplitude rounded on a 2^{-24} grid contributes less than 10^{-7} relative error at nominal $k = .2$; it fits within η_u and uses 24 more label bits. Eq. (A4) gives $1 - \xi < 4.42935 \times 10^{-7}$ and $k_{\mathrm{eff}}/k \in [.999030268, 1.000970203]$. For $\alpha = 1.1$, $h = .5$, $N = 4096$ probes at each of $\pm h$ and nominal $k = .2$, the calibration's collection-dependent dephasing error is below .000946. This contrasts with the loss of content at $\xi = .99$ above; it does not include parity-gate or detector errors.

A second numerical representation replaces frequency multiplication by cell midpoints while retaining g itself. The invariant vector in each cell is $g\mathbf{1}_{\mathrm{cell}}/\|g\mathbf{1}_{\mathrm{cell}}\|$, with coupling $\sqrt{\int_{\mathrm{cell}} J}$. Its finite matrix approximates the continuum propagator with operator error at most TB/M for M cells. These cells are a numerical resolution, not fresh physical ports. Dense and sparse matrix calculations agree with the moment enclosure and the prepared waveform. The recurrence and rounding rule specify the waveform without a fitted infinite-precision function.

After an encounter, swapping c with an independent same-frequency vacuum oscillator returns c exactly to vacuum, even when its old state is correlated with a reference. The swap exports E_c and has the finite duration and occupation account of Proposition 21. It supplies a reusable receiver for a finite bank of encounters, not a free thermal sink. Repeating the query law needs independent field preparations as well as independent vacuum exhausts. Independent field sectors selected one at a time give a finite-bank realization whose Hamiltonian remains bounded below. 8192 queries consume 8192 such preparations and exhausts, while sharing the waveform program. In this calibration example the nominal source energy is $E_{\mathrm{probes}}/[\hbar(2\pi J_0)] = 4096(.2)(20)[(1 - .55)^2 + (1 + .55)^2] = 42680.32$. It is predominantly collected and later exported, not eliminated by the small content-damage bound. Serial preparation and capture take $8192(2T) = 262144$ time units before parity control and receiver reset. Time windows in one band-limited waveguide are not automatically independent; that alternative requires a return/reset bound. Thermal exhaust resets c to a thermal state and changes the readout law. For example, for the exactly matched mode, a thermal receiver of mean occupation n_T gives even-parity probability

$$\frac{1}{2}\left[1 + \frac{e^{-2\xi k m^2/[1+2(1-\xi)n_T]}}{1 + 2(1-\xi)n_T}\right]$$

after mixing with the vacuum source packet, rather than the vacuum formula. For an unmatched packet, the thermal occupation term instead uses $|\langle c|S|c\rangle|^2 n_T$, which need not equal $(1-\xi)n_T$.

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